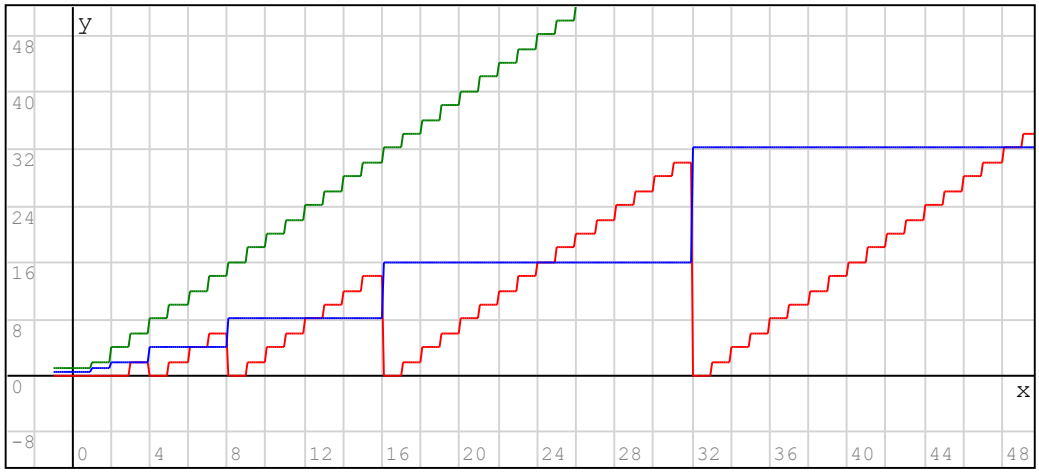
Created using a free version of SMath Studio

$y(x) := \sqrt{x}$

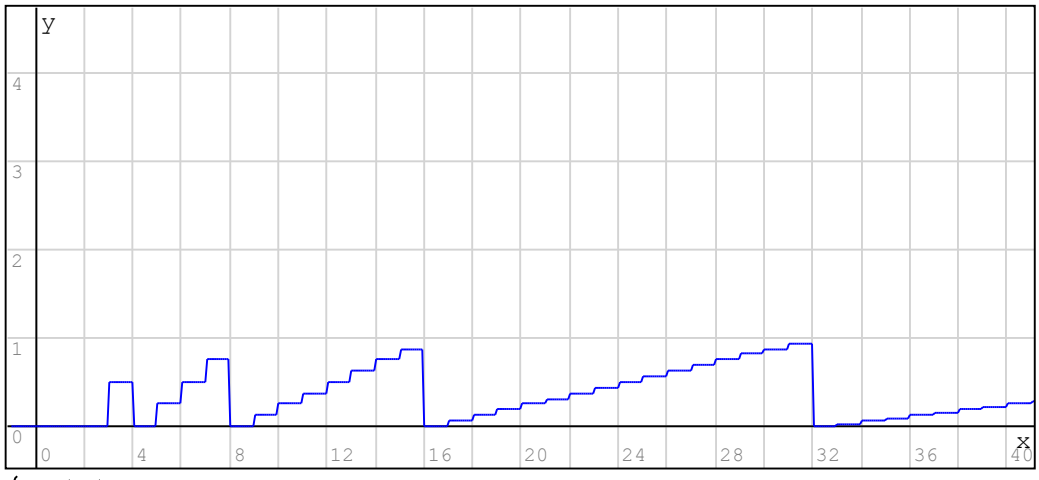
$clz(x) := \text{if } x = 0 \; 32 \text{ else } 32 - \text{trunc}\left(\log_2\left(\text{trunc}(x) + \frac{2^{32}-1}{2^{32}}\right) + \frac{2^{32}-1}{2^{32}}\right)$

$msb(x) := 32 - clz(x)$

$dX(x) := \frac{\text{trunc}\left(\text{trunc}(x) - 2^{msb(x)-1}\right)}{2^{msb(x)-1}}$



$$\begin{cases} 2^{msb(x)-1} \\ dX(x) \cdot 2^{msb(x)} \\ 2^{msb(x)} + dX(x) \cdot 2^{msb(x)} \end{cases}$$



$\{dX(x)$

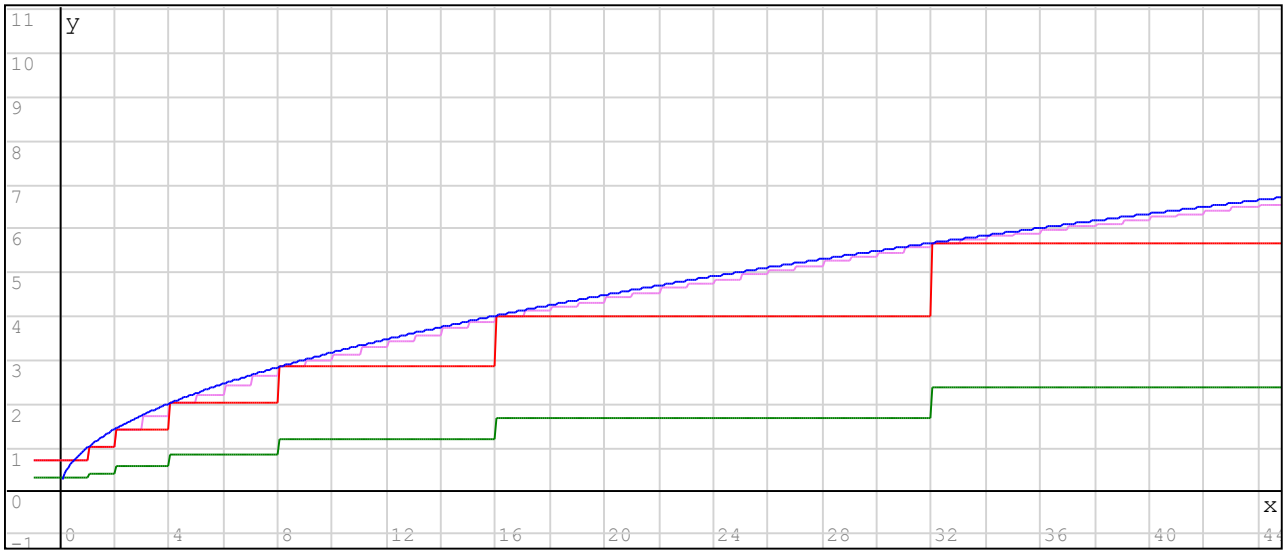
Not for commercial use

Created using a free version of SMath Studio

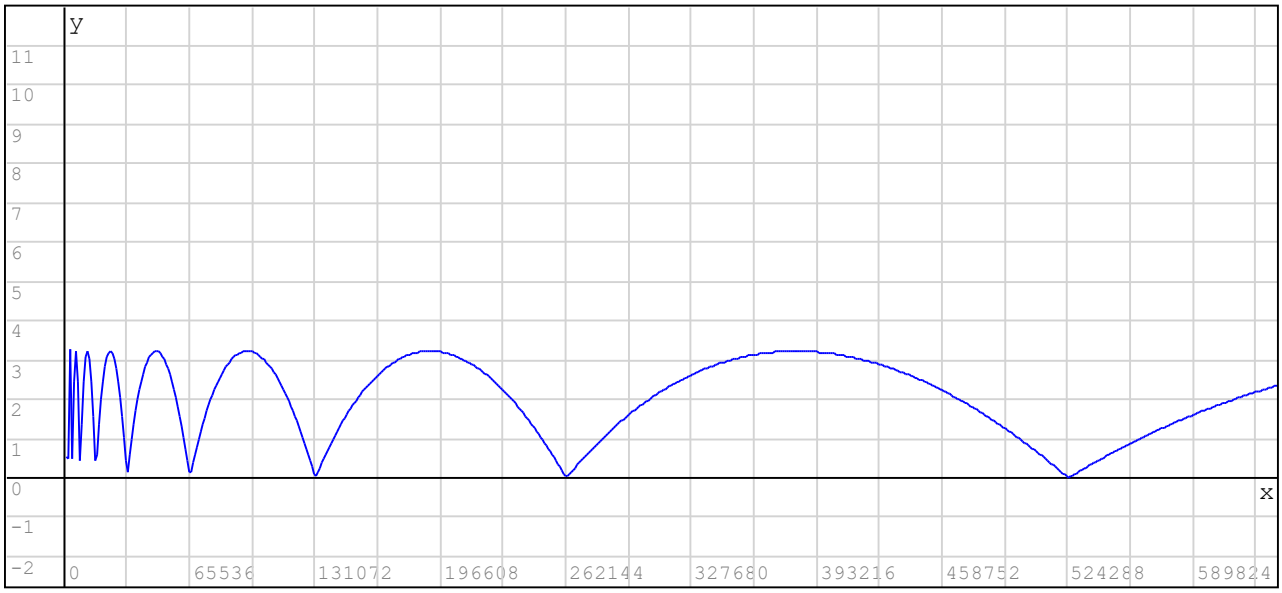
$$sqrtLinearY0(x) := \sqrt{2}^{msb(x)-1}$$

$$sqrtLinearDY(x) := \left(\sqrt{2}\right)^{msb(x)} - \left(\sqrt{2}\right)^{msb(x)-1}$$

$$sqrtLinear(x) := sqrtLinearY0(x) + \left(\frac{sqrtLinearDY(x)\cdot \mathrm{trunc}\left(dX(x)\cdot 2^{32}\right)}{2^{32}}\right)$$



$$\begin{cases} \sqrt{x} \\ sqrtLinearY0(x) \\ sqrtLinearDY(x) \\ sqrtLinear(x) \end{cases}$$



$$\frac{\sqrt{x} - sqrtLinear(x)}{\left(\sqrt{2}\right)^{msb(x)-16}}$$

Not for commercial use